# E-COMMERCE SALES INTELLIGENCE

## A Revenue Operations Case Study

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Analyst: Miracle Ezekiel

Tools: Google Sheets MySQL Workbench Python GitHub

Dataset: 5,000 Transactions October 2023 to October 2025

Repository: github.com/miracleezekiel/ecommerce-sales-intelligence

# 1. Executive Summary

This case study documents a full end-to-end data analytics project built against a synthetic e-commerce dataset of 5,000 transactions and 4,844 unique customers spanning October 2023 to October 2025.

The analysis reveals a business generating 533,666,024.35 in total revenue while facing two structural problems that are silently eroding its long-term commercial health.

**Problem 1 | The Retention Crisis**

96.82% of all customers placed exactly one order and never returned. The business is entirely dependent on constant new customer acquisition to sustain revenue. There is no returning customer base.

**Problem 2 | The Discount Trap**

Every discount tier from 5% to 20% reduces average order value compared to zero-discount transactions. The discount strategy is systematically trading profit for activity.

Using Google Sheets for data cleaning and exploration, MySQL Workbench for SQL analysis, and Python for RFM customer segmentation, all 4,844 customers have been scored, classified, and prioritised for targeted CRM action.

The primary recommendation is an immediate automated re-engagement programme targeting 1,543 Promising customers and 175 At Risk High Value customers before they drift into permanent churn.

| Metric | Value |
|---|---|
| Total Revenue | 533,666,024.35 |
| Total Profit | 79,708,734.91 |
| Average Order Value | 106,733.20 |
| Average Profit Margin | 14.92% |
| Unique Customers | 4,844 |
| One-Time Customers | 4,690 (96.82%) |
| Repeat Customers | 154 (3.18%) |

| Metric | Value |
|---|---|
| RFM Segments Identified | 5 |

# 2. Business Context

## 2.1 The Surface Picture

On first inspection this business looks healthy. Revenue is spread across four regions. Ten product categories perform with reasonable consistency. Monthly trends show no dramatic collapses. Average profit margin sits at 14.92% which is commercially reasonable for an e-commerce operation.

## 2.2 What the Data Actually Reveals

Beneath the headline numbers the data tells a different story entirely. Revenue is being sustained by a constant stream of new customers replacing the ones who quietly disappear after their first purchase. This is not growth. This is a business running in place while burning acquisition budget to stay there.

There is no loyalty engine. There is no automated retention system. There is no mechanism watching for customers who are about to leave. The business does not know it is losing 96.82% of its customers because the revenue line keeps moving upward as new customers arrive to fill the gap left by the ones who left.

## 2.3 Why Customer Retention Matters

Research consistently shows that acquiring a new customer costs five to seven times more than retaining an existing one. A business where nearly 97 out of every 100 customers never return is spending the maximum possible amount on acquisition while capturing none of the compounding value that loyal customers generate.

The data in this project makes this concrete. Customers who placed two orders spent 2.03 times more per lifetime than one-time customers. Customers who placed three orders spent 3.88 times more. Every additional order a customer places compounds their total value to the business. And almost nobody came back for a second order.

**The Business Case in One Number**

If just 10% of the 1,543 Promising customers could be converted into two-order customers, the business would add approximately 15,000 additional repeat buyers to its base. Each spending twice the average. With no additional acquisition cost.

# 3. Data Overview

## 3.1 Dataset Specifications

| Detail | Value |
| --- | --- |
| Total Transactions | 5,000 |
| Original Columns | 14 |
| Columns After Cleaning | 19 (5 calculated columns added) |
| Date Range | October 2023 to October 2025 |
| Unique Customers | 4,844 |
| Regions | 4 (North, East, West, South) |
| Cities | 20 (5 per region) |
| Product Categories | 10 |
| Payment Modes | 5 |
| Discount Levels | 5 (0%, 5%, 10%, 15%, 20%) |

## 3.2 Data Validation Checks

Before any analysis began the raw dataset was subjected to five structured validation checks in Google Sheets. This step protected every insight that followed.

| Check | Method | Finding |
| --- | --- | --- |
| Missing Values | COUNTBLANK across all columns | Zero missing values across all 14 columns |
| Duplicate Records | COUNTIF on Order ID | Zero duplicates all 5,000 Order IDs unique |
| Categorical Consistency | COUNTIF on Region, Category, Payment Mode | All values confirmed within expected ranges |
| Sales Formula Accuracy | Mathematical verification against Quantity x Price x Discount | 100% accuracy across all 5,000 rows |
| Profit Outlier Detection | Interquartile Range method | 223 transactions flagged above upper bound of 48,831.53 |

## 3.3 Calculated Columns Added

| Column | Formula Type | Purpose |
|---|---|---|
| Outlier_Flag | IF formula with IQR threshold | Flags 223 high-profit transactions without removing them |
| Profit_Margin_Pct | Profit divided by Sales multiplied by 100 | Enables margin comparison across all dimensions |
| Year | YEAR function on Order Date | Enables year-over-year analysis |
| Month | MONTH function on Order Date | Enables monthly trend analysis |
| Month_Name | TEXT function on Order Date | Human-readable month labels for charts |

## 3.4 Documented Data Issues

| Issue | Detail | Treatment |
|---|---|---|
| ISSUE-001 | 223 high-profit outlier transactions above IQR upper bound | Flagged and retained  not deleted |
| ISSUE-002 | 2023 data covers October to December only  partial year | Excluded from year-over-year comparisons |
| ISSUE-003 | Synthetic product names  not representative of real products | Analysis conducted at Category and Sub-Category level only |
| ISSUE-004 | Repeat purchase rate below 5% | Documented as central focus of Phase 4 CRM analysis |

# 4. Key Findings

## 4.1 Overall Revenue and Profitability

| Metric | Value | Significance |
|---|---|---|
| Total Revenue | 533,666,024.35 | Verified across Google Sheets, SQL, and Python |
| Total Profit | 79,708,734.91 | 14.92% average margin |
| Average Order Value | 106,733.20 | Baseline for all segment comparisons |
| Strongest Month | May  9.51% of annual revenue | Consistent across both 2024 and 2025 |
| Weakest Month | February  7.45% of annual revenue | Only 2.06% spread between strongest and weakest |

## 4.2 Regional Performance

| Region | Total Revenue | Margin | Key Finding |
|---|---|---|---|
| North | 143,578,246.10 | 14.84% | Leads revenue but has weakest margin and worst retention at 99.61% churn |
| East | 135,811,637.95 | 14.91% | Consistent across all metrics  no extremes |
| West | 131,045,973.35 | 14.97% | Weakest bottom category margins |
| South | 123,230,166.95 | 14.98% | Lowest revenue but best retention at 0.75% repeat rate |

## 4.3 The Discount Trap Confirmed

Across all 5,000 transactions higher discounts consistently produce lower average order values and lower total profit. This pattern holds without exception at every discount tier.

| Discount Level | Average Order Value | Change vs Zero Discount |
|---|---|---|
| 0% | 117,340.72 | Baseline  highest value |
| 5% | Declining | Below baseline |

| Discount Level | Average Order Value | Change vs Zero Discount |
|---|---|---|
| 10% | Declining | Below baseline |
| 15% | Declining | Below baseline |
| 20% | 94,618.46 | 19.4% below zero discount baseline |

**The Discount Trap in Plain English**

More discount. More orders. Less money. The business is running promotions that generate transaction volume while actively destroying profit per sale. This is confirmed across all 5,000 rows in SQL.

## 4.4 The Retention Crisis

| Metric | Value | Business Meaning |
|---|---|---|
| One-Time Customers | 4,690 (96.82%) | Nearly all customers disappear after one purchase |
| Two-Order Customers | 152 (3.14%) | Two-order customers spend 2.03x more per lifetime |
| Three-Order Customers | 2 (0.04%) | Three-order customers spend 3.88x more per lifetime |
| Best Regional Retention | South 0.75% | Best performing region still critically low |
| Worst Regional Retention | North 0.39% | Only 5 repeat customers out of 1,283 total |

## 4.5 Most and Least Profitable Combinations

| Rank | Region and Category | Average Margin | Action |
|---|---|---|---|
| 1 of 40 | North Electronics | 16.23% | Increase investment and marketing priority |
| 2 of 40 | East Clothing | 15.85% | Protect margin review discount policy |
| 39 of 40 | North Books | 13.89% | Review pricing and discount levels |
| 40 of 40 | North Beauty | 13.28% | Lowest margin in dataset pricing review urgent |

### 4.6 Outlier Transaction Impact

| Group | Transactions | Revenue Share | Profit Share | Avg Order Value |
|---|---|---|---|---|
| Outlier Flagged | 223 (4.46%) | 11.78% | 16.81% | 282,012.98 |
| Non-Outlier | 4,777 (95.54%) | 88.22% | 83.19% | 98,550.79 |

The 223 outlier transactions generate 16.81% of total profit from only 4.46% of all transactions. Removing them drops average profit per order from 15,941.75 to 13,880.37 a 13.0% reduction. Every performance benchmark must account for this inflation effect.

### 4.7 RFM Customer Segmentation Results

| Segment | Customers | Revenue | Avg Order Value | CRM Priority |
|---|---|---|---|---|
| Lost Customer | 2,972 (61.35%) | 53.46% | 95,997.47 | 5 Re-engage |
| Promising | 1,543 (31.85%) | 31.05% | 107,382.27 | 4 Nurture |
| At Risk High Value | 175 (3.61%) | 9.16% | 279,303.37 | 1 Urgent |
| Loyal Customer | 152 (3.14%) | 6.18% | 108,439.45 | 3 Reward |
| Champion | 2 (0.04%) | 0.15% | 137,836.50 | 2 Protect |

## 5. CRM Recommendations

All five recommendations below are directly supported by findings from the data. Each one identifies a specific customer group, explains the urgency, and describes a concrete action.

**Priority 1 | URGENT | At Risk High Value Customers | 175 Customers**

These 175 customers averaged 279,303.37 per order nearly three times the dataset average. They spent significantly, trusted the brand, and then went completely silent. Every additional month without contact reduces re-engagement probability. Personalised outreach within the next 30 days. Reference their previous category purchase. Offer a category-specific loyalty incentive not a blanket discount.

**Priority 2 | NURTURE | Promising Customers | 1,543 Customers**

1,543 customers bought recently and are still within the re-engagement window. The probability of a second purchase drops sharply after 30 days of post-purchase silence. Automated 3-step email sequence: Day 3 thank you with related product recommendations. Day 14 educational content related to purchase category. Day 30 loyalty programme invitation with first-return incentive.

**Priority 3 | REWARD | Loyal Customers | 152 Customers**

The 152 customers who already came back at least once are the most important people in this database. Their average profit margin of 15.47% is the highest of any segment. They buy less on discount and more at full price because they trust the brand. A formal loyalty programme with points, early access, and quarterly recognition will keep them before a competitor does.

**Priority 4 | POLICY | Discount Strategy Review | Business Wide**

The discount trap is confirmed across all 5,000 transactions. Reserve discounts exclusively for win-back campaigns targeting At Risk and Lost segments not as a general revenue driver. Test a zero-discount period for North Electronics and East Clothing the two highest margin combinations and measure impact on volume and profit simultaneously.

**Priority 5 | PILOT | North Region Retention Focus | 1,283 Customers**

North generates the most revenue but has the worst retention rate at 99.61% churn. It is the most acquisition-dependent region. Run the Promising customer nurture sequence as a North-specific pilot. Use South as the control region. Measure re-engagement rate at 30, 60, and 90 days. Even recovering 2% of churned customers in North would represent meaningful incremental revenue.

# 6. Technical Summary

## 6.1 Project Phases and Tools

| Phase | Tool | What Was Built |
|---|---|---|
| Phase 1  Data Cleaning | Google Sheets | 5 validation checks  4 documented issues  5 calculated columns  19 total columns |
| Phase 2  Exploratory Analysis | Google Sheets | 15 KPIs  5 pivot tables  5 charts  6 analytical sections |
| Phase 3  SQL Analysis | MySQL Workbench 8.0 CE | 15 KPI queries  4 segmentation queries  3 churn analysis queries |
| Phase 4  CRM Analysis | Python in Cursor | 13 notebook cells  RFM scoring segment chart  4,844 row customer export |
| Phase 5  Case Study | GitHub  WPS Document | Full case study published  repo made public  LinkedIn and Twitter content released |

## 6.2 SQL Files and Query Summary

| File | Queries | Topics Covered |
|---|---|---|
| 03_kpi_queries.sql | 15 | Overall performance  regional category  discount  time trends payment  margin  customer outlier |
| 04_segmentation_queries.sql | 4 | City performance  customer frequency  transaction value tiers  discount by category |
| 05_churn_analysis.sql | 3 | Inactive customer identification purchase gap analysis category churn risk |

## 6.3 Python Notebook Summary

| Cell | Title | What It Did |
|---|---|---|
| 1 | Import Libraries | Loaded pandas  matplotlib seaborn |

| Cell | Title | What It Did |
|---|---|---|
| 2 | Load Dataset | Confirmed 5,000 rows and 19 columns |
| 3 | Preview Data | First 5 rows verified correct |
| 4 | Column Names | All 19 columns confirmed |
| 5 | Data Types | Zero nulls confirmed  types verified |
| 6 | Statistical Summary | Figures matched Phase 2 and Phase 3 exactly |
| 7 | Date Conversion | Order_Date converted from text to datetime64 |
| 8 | Build RFM Table | Recency, Frequency, Monetary calculated for all 4,844 customers |
| 9 | Assign RFM Scores | 1 to 3 score assigned per customer per dimension |
| 10 | Assign Segments | Five segment labels assigned to all 4,844 customers |
| 11 | Visualise Segments | Professional bar chart created and saved |
| 12 | Revenue Analysis | Revenue and margin calculated per segment |
| 13 | Export Data | rfm_customer_segments.csv exported  4,844 rows  10 columns |

## 6.4 Phase 4 Deliverables

| File | Description |
|---|---|
| rfm_segment_chart.png | Bar chart showing customer count and percentage per CRM segment |
| rfm_customer_segments.csv | All 4,844 customers with RFM scores  segment labels  and CRM priority ranking |

# 7. Conclusion

This analysis began with 5,000 transactions and a question nobody had asked directly.

Is this business actually retaining its customers?

The answer is no. Emphatically, consistently, and business-wide, no.

96.82% of customers left after one purchase. The revenue numbers looked healthy because new customers kept arriving to fill the gap. But a business that cannot retain customers is not growing. It is running in place while burning acquisition budget to stay there.

The data also reveals the opportunity on the other side of that finding. The customers who did return spent twice as much per lifetime. The customers who returned twice spent nearly four times as much. Every additional order a customer places compounds their value to the business.

The CRM system required to capture that compounding value does not need to be complex. It needs to be intentional. A structured nurture sequence for Promising customers. A personalised win-back campaign for At Risk High Value customers. A formal loyalty programme for the 152 customers who already came back once.

These are not expensive interventions. They are automated, targeted, and directly supported by two years of transaction evidence.

The data has already done the work of identifying who needs attention and in what order. The only remaining question is whether the business acts on it.

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Full project repository including all datasets, SQL queries, Python notebooks, and documentation:
**github.com/miracleezekiel/ecommerce-sales-intelligence**

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